

Track 37: CAD/CAM & Mechanical Design

June 2026

Track 37: CAD/CAM & Mechanical Design

Department: Mechanical Engineering / Mechatronics | MR ROKESH TECHNOLOGY

Skill path: Engineering Drawing → SolidWorks/Fusion 360 → Assembly → Simulation → Drafting → CAM Toolpaths

Related: [Track 42 — Mechatronics Integration](#) | [Department Map](#)

Track Overview

Who this is for: Mechanical and mechatronics students targeting design engineer and manufacturing roles.

Prerequisites: Basic engineering mechanics and drawing standards.

Tools: SolidWorks (or Fusion 360), ANSYS Mechanical (intro), CAM module, 3D printing lab (optional).

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	3D modeling awareness	Part + assembly model
1 Month	80–100	Junior design engineer	Mechanism assembly + drawings
3 Months	250–300	Internship-ready	Designed + simulated component
6 Months	500–600	Design engineer ready	Portfolio of 5+ mechanical designs

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Sketch constraints, extrude, revolve, patterns	5 part models
D3–4	Assembly mates, sub-assemblies, exploded views	Mechanism assembly
D5–6	Engineering drawings: GD&T intro, BOM	Manufacturing drawing set
D7–8	Simulation: static stress, factor of safety	FEA report (simple part)
D9–10	CAM intro, capstone	Designed part + drawing + sim report

1-Month Foundation

Week	Topics	Project
W1	Advanced modeling: sweeps, lofts, surfaces	Complex part library
W2	Top-down vs bottom-up assembly design	Gear/ linkage mechanism
W3	Sheet metal, weldments, configurations	Fabrication-ready design
W4	Simulation + technical drawings	Complete design project

3-Month Job Ready

Month	Topics	Projects
M1	Machine elements design (shafts, bearings, gears)	Gearbox assembly
M2	FEA: thermal, modal analysis intro	Optimized bracket design
M3	CAM toolpaths, capstone	Design-to-manufacture capstone

6-Month Professional

M1–M3 + advanced simulation + design portfolio + industry standard parts (fasteners library) + 3D print prototype.

Assessment Rubric

Weekly Quiz 20% | CAD labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Enforce drawing standards (IS/ASME) from Week 1. Pair with [Track 38](#) for control system simulation.